Finding the Bus's Distance from its Route

The 'fast formula' explained, with worked numbers

The route is a line on the map; the bus is somewhere off it. We want the perpendicular distance — the shortest gap between the bus and the road.

Step 1 — Pretend the Earth is flat

Over a few kilometres, Earth's curvature is invisible. So we convert each latitude/longitude into plain (x, y) metres, measured from a reference point near the route:

$$x = R \cdot (\lambda - \lambda_0) \cdot \cos \varphi_0 \qquad \text{(metres east of reference)}$$

$$y = R \cdot (\varphi - \varphi_0) \qquad \text{(metres north of reference)}$$

R = Earth's radius, 6,371,000 m. The $\cos \varphi_0$ corrects for the fact that a degree of longitude is shorter away from the equator. After this conversion, every point is just (x, y) in metres and we can use regular geometry.

Step 2 — Example

Reference point = Stop P_0 (so P_0 itself sits at the origin).

Point	x (m)	y (m)	What it means
P_0	0	0	The reference itself
P_1	–1192	+1133	1192 m west, 1133 m north of P_0
Bus 17	–613	+556	613 m west, 556 m north of P_0

Distance from P_0 to P_1: $\sqrt{1192^2 + 1133^2} \approx 1645$ m.

Step 3 — Find the closest point on the segment to the bus

From P_0, draw two arrows:

- **Arrow S** = P_0 to P_1 (along the road) = (–1192, +1133)
- **Arrow V** = P_0 to bus = (–613, +556)

The fraction of the way along the segment that gets you closest to the bus is:

$$t^\star = \frac{\vec{V} \cdot \vec{S}}{\|\vec{S}\|^2}$$

where $\vec{V} \cdot \vec{S}$ is the dot product (multiply matching components, then add) and $\|\vec{S}\|^2$ is the segment length squared. **In plain English: how much of arrow V points the same direction as arrow S.**

Numbers:

- $\vec{V} \cdot \vec{S} = (-613)(-1192) + (556)(1133) = 730\,696 + 629\,948 = 1\,360\,644$
- $\|\vec{S}\|^2 = 1192^2 + 1133^2 = 2\,704\,553$
- $t^\star = 1\,360\,644 \,/\, 2\,704\,553 \approx$ **0.503**

So the bus is **50.3% of the way along the segment** — basically at the midpoint.

The `clip(…, 0, 1)` step in the original formula just snaps to an endpoint if the bus is past either end of the segment. With $t^\star = 0.503$ we're already inside [0, 1], so no clipping happens.

Step 4 — Where is that closest point, and how far is the bus from it?

$$\text{closest point} = P_0 + 0.503 \times (P_1 - P_0) = (-600,\ +570)$$

$$\text{offset} = (-613 - (-600),\ 556 - 570) = (-13,\ -14)$$

$$\delta = \sqrt{13^2 + 14^2} = \sqrt{365} \approx \textbf{19 m}$$

Step 5 — Verdict

Bus 17 is **19 m off the centre line** of its route. Our corridor is 50 m wide.

> 19 ≤ 50, so the bus is on route. Carry on.

The picture

```
P₀ ─────────────── midpoint ─────────────── P₁
                      │
                      │  19 m  (perpendicular drop)
                      │
                   Bus 17
```

The dot-product trick slides the bus's position sideways onto the road, then measures the leftover gap. That gap is δ. If $\delta \le 50$ m the bus is on route; if it exceeds 50 m for **three consecutive packets**, we raise an `OFF_ROUTE` alert (and the divert-alert pipeline fires).